Metadata: StormSense Comma Separated Values (CSV) for 30-Day Hindcasts (30day)

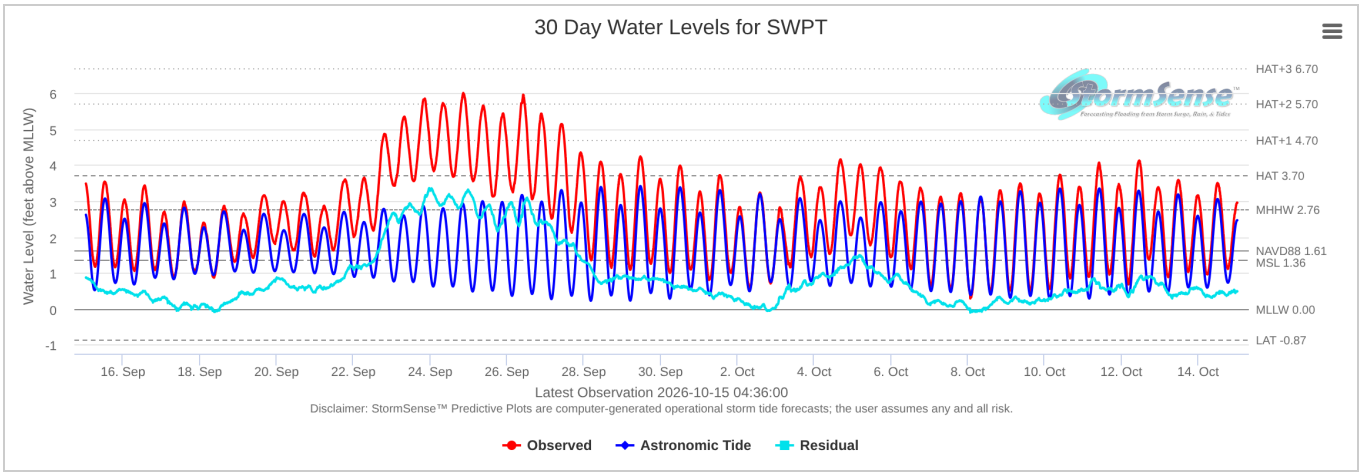


Figure 1. *Example StormSense Chart of 30 days of hindcast data (MLLW) at NOAA's Sewells Point tide gauge, in Norfolk, VA, looking backward 30 days from October 15, 2026 at 16:36 EDT (2026-10-15T20:36:00Z).*

Fields **30day data downloads only include observations from the last 30 days hindcast period**

date_time — Date and time in Zulu/GMT, formatted as a string leading with the date as "YYYY-MM-DD" followed by the time (T) as "hh:mm:ss" relative to Zulu (Z) time. Recorded in regular 6-minute intervals (0, 6, 12 … 54 minutes past the hour), listed in chronological order.

water_level_ft — (**Observed**) Observed storm tide water level for the past 30 days at this monitoring location. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).

astro_tide — (**Astronomic Tide**) Harmonically predicted astronomical tide at the same times, from P-ETSS (NOAA CO-OPS tide predictions further back at NOAA gauges, which match P-ETSS's tide to 0.01 ft). Blank where no prediction reaches back that far. Formatted in units of feet rounded to three decimal places (thousandth of a foot) relative to Mean Lower Low Water (MLLW).

residual — (**Residual**) Observed residual: water_level_ft – astro_tide, the non-tidal water level from wind, air pressure, rain and river flow. Feet; the same in MLLW and NAVD88. Blank where astro_tide is blank.

Table 1. *Example StormSense CSV Table download of 30day hindcast data (MLLW) at NOAA's Sewells Point tide gauge, in Norfolk, VA, through October 15, 2026 at 16:36 EDT (2026-10-15T20:36:00Z).*

date_time	water_level_ft	astro_tide	residual
2026-09-15T20:48:00Z	1.859	1.412	0.447
2026-09-15T20:54:00Z	1.820	1.360	0.460
2026-09-15T21:00:00Z	1.764	1.309	0.455
2026-09-15T21:06:00Z	1.708	1.260	0.449
2026-09-15T21:12:00Z	1.656	1.213	0.444
2026-09-15T21:18:00Z	1.590	1.167	0.423
2026-09-15T21:24:00Z	1.558	1.124	0.434
2026-09-15T21:30:00Z	1.505	1.082	0.423
2026-09-15T21:36:00Z	1.469	1.043	0.426
2026-09-15T21:42:00Z	1.426	1.005	0.421
…	…	…	…
2026-10-15T20:18:00Z	2.762	1.763	0.999
2026-10-15T20:24:00Z	2.719	1.711	1.009
2026-10-15T20:30:00Z	2.660	1.660	1.000
2026-10-15T20:36:00Z	2.611	1.610	1.001

Datums *(gray lines labeled at the right of the chart; click a label to hide or show its line)*

····· HAT+1 … HAT+4	Reference lines 1, 2, 3 and 4 feet above HAT, for reading how far an extreme water level rises above the highest predicted tide. Where a station has no published HAT, the steps start from MHHW.
– – – HAT	Highest Astronomical Tide: the highest water level predicted from the tide alone (no weather) over the National Tidal Datum Epoch (NTDE, 1983–2001).
- - - - MHHW	Mean Higher High Water: the average of the higher of each day's two high tides over the NTDE; a common threshold for nuisance (high-tide) flooding.
—— NAVD88	North American Vertical Datum of 1988: the fixed, land-based geodetic datum used for surveyed elevations, roads and flood maps.
– – – MSL	Mean Sea Level: the average of hourly water levels over the NTDE.
—— MLLW	Mean Lower Low Water: the average of the lower of each day's two low tides over the NTDE. It is zero on these charts and in these CSVs.
– – – LAT	Lowest Astronomical Tide: the lowest water level predicted from the tide alone over the NTDE.

Datum source: at NOAA-operated tide gauges the datums are NOAA's published tidal datums for that station. USGS, VDEM and StormSense sensors natively report live water levels in NAVD88 only, so their MLLW forecasts and observations are datum-shifted from NAVD88 using the MLLW–NAVD88 separation derived from NOAA VDatum (https://vdatum.noaa.gov/vdatumweb/), and their tidal datum lines come from VDatum as well. Not every datum is available at every station.